Symmetries in Physics — Exercise Sheet 6

Symmetry, projectors, and degeneracy

Variant: Questions only

Exercise 1. In Lecture 6's example of a particle hopping on the corners of a triangle, $C_3 = \langle r \rangle$ acts on the orbitals by

$$\rho(r)\,|j\rangle = |j+1\rangle\,,$$

with indices taken modulo 3. This is the regular representation of C_3 (Sheet 3, Exercise 5). Lecture 6's self-study section obtains the vectors $v_k = \sum_{j=1}^{3} \omega^{-kj}\,|j\rangle$, where $\omega = e^{2\pi \mathrm{i}/3}$, by applying the character projectors to $|3\rangle$. Fill in the details, starting from $|1\rangle$.

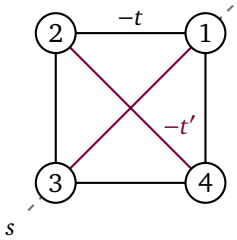
(i) Write down the three character projectors P_k of C_3.

(ii) Apply each of them to $|1\rangle$, and compare the result with v_k.

(iii) Check that $P_0 + P_1 + P_2 = \mathbb{1}$, and explain why the image of each P_k is a line.

Exercise 2 (retrieval: parity through character projectors). Return to the parity action of $C_2 = \{e, p\}$ on $L^2(\mathbb{R})$ from Sheet 3, Exercise 4: $(\Pi\psi)(x) = \psi(-x)$, where $\Pi = \rho(p)$. Using the two characters and the character-projector formula of Lecture 6, compute both projectors and identify their images. The formula was proved for finite-dimensional spaces, but for a group of order 2 the averaging makes sense on $L^2(\mathbb{R})$ directly, and you may use it in that form. Explain why each image is an isotypic component but is not itself an individual irreducible representation.

Exercise 3. Four identical orbitals $|1\rangle, \ldots, |4\rangle$ sit at the corners of a square, numbered anticlockwise. An electron hops between neighbouring corners with amplitude $-t$ and across a diagonal with amplitude $-t'$, where $t > 0$ and t' is real. The symmetry group is D_4: the rotation r through $\pi/2$ acts by $|j\rangle \mapsto |j+1\rangle$ (indices modulo 4), and s is the mirror in the diagonal through corners 1 and 3 (dashed). Take the orbitals orthonormal, with on-site energy zero.

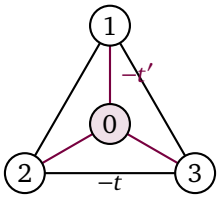


(i) With the classes of Lecture 5's D_4 table, $\{e\}$, $\{r^2\}$, $\{r, r^3\}$, $\{s, r^2 s\}$, $\{rs, r^3 s\}$, show that the permutation character is $\chi_{\mathrm{perm}} = (4, 0, 0, 2, 0)$ and that $\mathbb{C}^4 \cong V^{(1)} \oplus V^{(3)} \oplus V^{(5)}$, where $V^{(i)}$ is the irreducible representation with character χ_i in that table.

(ii) Apply the character projectors P_1, P_3 and P_5 to $|1\rangle$.

(iii) Every irreducible representation occurs once, so the projected vectors are eigenvectors of H. Find the three levels and their degeneracies. For which t' do two levels meet? Is that degeneracy accidental? (Look at H when $t' = t$; for $t' = -t$, first change the signs of $|2\rangle$ and $|4\rangle$.)

Exercise 4. Put a fourth atom at the centre of the triangle of Lecture 6, with one orbital $|0\rangle$ of energy ε. The corner orbitals and the hopping $-t$ between corners are as in the lecture, and the hopping between the centre and each corner is $-t'$, with $t > 0$, $t' \geq 0$ and ε real. The group C_{3v} fixes $|0\rangle$ and permutes the corners.



(i) Show that $\mathbb{C}^4 \cong 2V^{(A_1)} \oplus V^{(E)}$.

(ii) Take the orthonormal symmetry-adapted basis $w_{1,1} = |0\rangle$, $w_{1,2} = \frac{1}{\sqrt{3}}\big(|1\rangle + |2\rangle + |3\rangle\big)$ of the isotypic component V_{A_1} (one partner, two copies). Show that the reduced matrices are

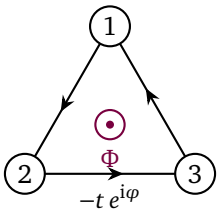
$$F^{(A_1)} = \begin{pmatrix} \varepsilon & -\sqrt{3}\,t' \\ -\sqrt{3}\,t' & -2t \end{pmatrix}, \qquad F^{(E)} = t.$$

(iii) Find the spectrum. Which degeneracy is enforced? For which t' is there a threefold level, and when is it accidental?

Exercise 5. Thread a magnetic flux through the triangle of Lecture 6. Its only effect on the hopping is a phase: an electron hopping anticlockwise, $1 \to 2 \to 3 \to 1$, picks up $e^{i\varphi}$, and one hopping clockwise $e^{-i\varphi}$. With $C\,|j\rangle = |j+1\rangle$ and $t > 0$,

$$H = -t\big(e^{i\varphi}\,C + e^{-i\varphi}\,C^{-1}\big).$$

For part (iii), first read the self-study passage *Keeping only C_3, and reality* in Section 6.5 of Lecture 6.



(i) Show that H is Hermitian and commutes with $\rho(r) = C$, but that for $\sin\varphi \neq 0$ it commutes neither with $\rho(s)$, for the mirror s of Lecture 6 that fixes corner 3 and swaps 1 and 2, nor with complex conjugation.

(ii) Show that Lecture 6's vectors $v_k = \sum_j \omega^{-kj}\,|j\rangle$ are still eigenvectors, with eigenvalues $\lambda_k = -2t\cos\big(\varphi + \frac{2\pi k}{3}\big)$.

(iii) Show that two levels coincide exactly when 3φ is a multiple of π. What has happened to the doublet E for other φ, and which of Lecture 6's two reasons for it has the flux removed?